

1. a) What is preference based approach of consumer's behavior? Explain and interpret the assumptions of preference based approach. *Chap-1* 1.1 → Pg 2-6

b) Define indifference set. Examine that if a preference relation satisfies strong monotonicity, its indifference curves must be downward sloping. Pg-16-17

✓ 2. a) Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity. *Chap-1* Pg-10-12

b) Show that,

$Strong Monotonicity \Rightarrow Monotonicity \Rightarrow LNS$ - Pg-13-15

✓ 3. a) What are the twin definitions of convexity? In what respect they are different from strong convexity-examine graphically. *Chap-1* Pg-17-20

b) Show that.

Pg-23-25

$Convexity\ of\ preferences \Leftrightarrow USC(x)\ is\ convex \Leftrightarrow u(.)\ is\ quasiconcave$

4. a) What are the typical utility functions used in economic applications? Write down their functional forms; also calculate and interpret the notions of MRS. *Chap-1* Pg-33-38

b) Prove that, if a preference relation satisfies monotonicity and continuity, then there exist a utility function $u(.)$ representing such preference relation. Pg-50-51

5. a) Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = R_+^N$, then examine the properties

satisfied by the Walrasian demand function $x(p, w)$. In what situations the sufficient condition of UMP may be violated. *Chap-2* Pg-78-85

b) Examine the following properties of preference relations:

i) Homogeneity

ii) Homotheticity

Pg-39-42 (first para)

6. Consider the Cobb-Douglas utility function:

$$u(x_1, x_2) = Kx_1^\alpha x_2^{1-\alpha} \quad \text{for some } \alpha \in (0,1) \text{ and } K > 0$$

- Math not done
- Derive the Walrasian demand functions and examine their properties.
 - How do you get the indirect utility functions? Explain their properties and also verify Roy's identity.
 - Solve the expenditure minimization problem and examine the characteristics of the compensated demand function.

7. a) Distinguish between compensating variation (CV) and equivalent variation (EV). In what respect they are different from the consumer's surplus (CS) Pg-165-166

- b) Illustrate that for a i) decrease in price of a commodity (when both goods are normal)

Chap-3

$$\text{Pg-168-172} \left[\begin{array}{l} \text{EV} > \text{CS} > \text{CV} - \text{Fig. 3.4+3.5} \\ \text{ii) for price increase } \text{EV} < \text{CS} < \text{CV} \rightarrow \text{Pg Fig. 3.6} \end{array} \right.$$

8. a) Define production sets. Graphically explain the properties of the production set. Pg 229-230

- Chap-4
- b) What is the elasticity of substitution (EoS). Find the elasticity of substitution for the production function $q = f(K, L) = aK + bL$. What will be the EoS if the production function is fixed proportions? Pg-246-247 - Example 4.5

9. a) Define profit function. If the production set Y is closed and have free disposal, then what properties do the profit function satisfies. - Pg-258-259

- Chap-1 - b) In case of well-defined utility function, show that Concavity $\nRightarrow$ Quasiconcavity

Pg-25-28

10. Consider a consumer with the utility function: $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{2}}$

- i) Derive the Walrasian and compensated demand functions.

- ii) Consider a) $\{p^0, m^0\} = \{4, 4, 300\}$

b) $\{p^1, m^0\} = \{2, 4, 300\}$

Compute the indirect utilities and costs.

- iii) Calculate EV, CV and CS. Based on their relationship examine the nature of the commodities.

Six sheet